

Fig. 3: Actual-size, single-side PCB of the letterbox

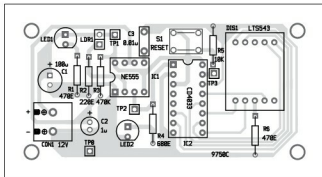


Fig. 4: Component layout of the letterbox

### Test Points

| Test point | Details                              |
|------------|--------------------------------------|
| TP0        | 0V, GND                              |
| TP1        | 12V                                  |
| TP2        | High when light is blocked from LDR1 |
| TP3        | High when S1 is pressed              |

and DIS1 increments the display by one count. That is, for each pulse received at input pin 1 of IC2, its

output advances by one count, which is reflected in the seven-segment display. When the counter reaches nine, it automatically resets to zero and the cycle repeats. Switch S1 is used to reset the counting.

After collecting the letter from the letterbox, always reset the counter using switch S1.

## Construction and testing

An actual-size, single-side PCB for the letterbox circuit is shown in

Fig. 3 and its component layout in Fig. 4. Enclose the PCB in a suitable box in such a way that light from LED1 falls on LDR1. Ensure proper wiring to avoid any mistake. For troubleshooting, check the voltages at various test points as listed in the table. ●

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